

第 2 周实验报告

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1 课堂实验

1.1 第 5 题

1.1.1 (1) 实验要求：

- 赋予脚本执行权限，在后台启动，并把 stdout、stderr 分别写入 stdout.log、stderr.log。
- 使用 Ctrl-Z 挂起任务，再用 bg 让它在后台继续；使用 jobs 确认状态。
- 使用 jobs -p 或等价方式取得 PID，不得手工抄写 PID；发送 SIGTERM 并等待任务结束。
- 确认 cleanup.log 中出现 CLEAN_EXIT，且任务已经退出；禁止使用 kill -9。

1.1.2 (2) 代码及运行结果：

```
#!/usr/bin/env bash
trap 'echo CLEAN_EXIT >> cleanup.log; exit 0' TERM INT
n=0
while true; do
    echo "$n"
    n=$((n+1))
    sleep 1
done
```

```
lumieredunord@lumieredunord-VirtualBox:~$ mkdir q05
lumieredunord@lumieredunord-VirtualBox:~$ cd q05
lumieredunord@lumieredunord-VirtualBox:~/q05$ nano worker.sh
lumieredunord@lumieredunord-VirtualBox:~/q05$ chmod +x worker.sh
lumieredunord@lumieredunord-VirtualBox:~/q05$ ./worker.sh > stdout.log 2> stderr.log
^Z
[1]+  已停止                  ./worker.sh > stdout.log 2> stderr.log
lumieredunord@lumieredunord-VirtualBox:~/q05$ bg
[1]+  ./worker.sh > stdout.log 2> stderr.log &
lumieredunord@lumieredunord-VirtualBox:~/q05$ jobs
[1]+  运行中                  ./worker.sh > stdout.log 2> stderr.log &
lumieredunord@lumieredunord-VirtualBox:~/q05$ jobs -p
2897
lumieredunord@lumieredunord-VirtualBox:~/q05$ kill $(jobs -p)
lumieredunord@lumieredunord-VirtualBox:~/q05$ cat cleanup.log
CLEAN_EXIT
[1]+  已完成                  ./worker.sh > stdout.log 2> stderr.log
```

1.2 第 6 题

1.2.1 (1) 实验要求:

- 在已配置好 Python、pytest 和 ruff 的课程环境中打开 q06，并启用 Python 语言服务器。
- 分别演示跳转到定义和查找引用。
- 使用“重命名符号”把 total_price 改为 calculate_total，保证定义和所有代码引用同步改变。
- 在 app.py 中临时加入未使用的 import os，用 ruff 发现并删除；最后运行 ruff 和 pytest 并保证通过。

1.2.2 (2) 代码及运行结果:

配置 python、pytest、ruff 环境

```
lumeredunord@lumeredunord-VirtualBox:~/q06$ curl https://bootstrap.pypa.io/pip/3.8/get-pip.py | sudo python3.8
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 2246k  100 2246k    0     0  1540k      0  0:00:01  0:00:01 --:--:-- 1541k
WARNING: The directory '/home/lumeredunord/.cache/pip' or its parent directory is not owned or is not writable by the current user. To fix this, you may have to delete the directory and create the directory again with a permissions of 0o700, and then run 'pip download --root-dir /home/lumeredunord/.cache/pip --quiet --no-input'.
Collecting pip<25.1
  Downloading pip-25.0.1-py3-none-any.whl.metadata (3.7 kB)
  Downloading pip-25.0.1-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 7.4 MB/s eta 0:00:00
WARNING: Error parsing dependencies of distro-info: Invalid version: '0.18ubuntu0.18.04.1'
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 9.0.1
    Uninstalling pip-9.0.1:
      Successfully uninstalled pip-9.0.1
  Successfully installed pip-25.0.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use pip in virtual environments.
lumeredunord@lumeredunord-VirtualBox:~/q06$ python3.8 -m pip install --user ruff pytest
Collecting ruff
  Downloading ruff-0.16.5-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (26 kB)
Requirement already satisfied: pytest in /usr/lib/python3/dist-packages (3.3.2)
Downloading ruff-0.16.5-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (10.3 MB)
    10.3/10.3 MB 6.3 MB/s eta 0:00:00
WARNING: Error parsing dependencies of distro-info: Invalid version: '0.18ubuntu0.18.04.1'
Installing collected packages: ruff
Successfully installed ruff-0.16.5
lumeredunord@lumeredunord-VirtualBox:~/q06$ python3.8 -m ruff --version
ruff 0.16.5
lumeredunord@lumeredunord-VirtualBox:~/q06$ python3.8 -m pytest --version
This is pytest version 3.3.2, imported from /usr/lib/python3/dist-packages/pytest.py
```

```
def total_price(price:float, count: int) -> float:
    return price*count
```

```
from math_utils import total_price
print(total_price(12.5, 4))
```

```
from math_utils import total_price
def test_total_price():
    assert total_price(12.5, 4) == 50.0
```

```

lumieredunord@lumieredunord-VirtualBox:~/q06$ nano math_utils.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ nano app.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ nano test_math_utils.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ python3.8 -m ruff check .
I001 [*] Import block is un-sorted or un-formatted
--> app.py:1:1
1 | / import os
2 | | from math_utils import calculate_price
3 | | print(calculate_price(12.5, 4))
help: Organize imports
1 | import os
2 | +
3 | from math_utils import calculate_price
4 | +
5 | print(calculate_price(12.5, 4))

F401 [*] `os` imported but unused
--> app.py:1:8
1 | import os
2 | | from math_utils import calculate_price
3 | | print(calculate_price(12.5, 4))
help: Remove unused import: `os`
1 | - import os
2 | | from math_utils import calculate_price

I001 [*] Import block is un-sorted or un-formatted
--> test_math_utils.py:1:1
1 | | from math_utils import calculate_price
2 | | def test_total_price():
3 | |     assert calculate_price(12,5, 4) == 50.0
help: Organize imports
1 | | from math_utils import calculate_price
2 | +
3 | +
4 | | def test_total_price():

Found 3 errors.
[*] 3 fixable with the `--fix` option.
lumieredunord@lumieredunord-VirtualBox:~/q06$ python3.8 -m ruff check . --fix
Found 3 errors (3 fixed, 0 remaining).
lumieredunord@lumieredunord-VirtualBox:~/q06$ python3.8 -m pytest

```

```

lumieredunord@lumieredunord-VirtualBox:~/q06$ nano math_utils.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ nano app.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ nano test_math_utils.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ python 3.8 -m ruff check . --fix
python: can't open file '3.8': [Errno 2] No such file or directory
lumieredunord@lumieredunord-VirtualBox:~/q06$ python3.8 -m ruff check . --fix
All checks passed!
lumieredunord@lumieredunord-VirtualBox:~/q06$ python3.8 -m pytest
===== test session starts =====
platform linux -- Python 3.8.0, pytest-3.3.2, py-1.5.2, pluggy-0.6.0
rootdir: /home/lumieredunord/q06, inifile:
collected 1 item

test_math_utils.py F

===== FAILURES =====
test_total_price

  def test_total_price():
    > assert calculate_total(12,5, 4) == 50.0
E   TypeError: calculate_total() takes 2 positional arguments but 3 were given

test_math_utils.py:5: TypeError
===== 1 failed in 0.01 seconds =====
lumieredunord@lumieredunord-VirtualBox:~/q06$ nano test_math_utils.py
lumieredunord@lumieredunord-VirtualBox:~/q06$ pytest3.8 -m pytest
pytest3.8: 未找到命令
lumieredunord@lumieredunord-VirtualBox:~/q06$ python3.8 -m pytest
===== test session starts =====
platform linux -- Python 3.8.0, pytest-3.3.2, py-1.5.2, pluggy-0.6.0
rootdir: /home/lumieredunord/q06, inifile:
collected 1 item

test_math_utils.py .

===== 1 passed in 0.01 seconds =====

```

1.3 第 7 题

1.3.1 (1) 实验要求:

- 先运行给定输入，确认结果错误或出现异常。
- 使用 pdb、IDE debugger 或等价调试器，在 merge 中设置断点并观察 i、j、left[i]、right[j]。
- 完成最小修复，不得用 sorted 替换归并排序。
- 使用 pytest 增加两个测试：给定输入和含重复元素的列表；保证测试通过。

1.3.2 (2) 代码及运行结果:

```
def merge(left, right):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i]); i += 1
        else:
            result.append(right[j]); j += 1
    return result + left[i:] + right[j:]

def merge_sort(a):
    if len(a) <= 1: return a
    m = len(a) // 2
    return merge(merge_sort(a[:m]), merge_sort(a[m:]))

print(merge_sort([3, 1, 4, 1, 5, 9, 2, 6]))
```

用 pdb 设置断点

```
lumieredunord@lumieredunord-VirtualBox:~$ mkdir q07
lumieredunord@lumieredunord-VirtualBox:~$ cd q07
lumieredunord@lumieredunord-VirtualBox:~/q07$ nano merge_sort.py
lumieredunord@lumieredunord-VirtualBox:~/q07$ python3.8 merge_sort.py
[1, 5, 4, 3, 4, 5, 6, 9]
```

```

def merge(left, right):
    import pdb; pdb.set_trace()
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i]); i += 1
        else:
            result.append(right[j]); j += 1
    return result + left[i:] + right[j:]

def merge_sort(a):
    if len(a) <= 1: return a
    m = len(a) // 2
    return merge(merge_sort(a[:m]), merge_sort(a[m:]))

print(merge_sort([3, 1, 4, 1, 5, 9, 2, 6]))

```

```

lumieredunord@lumieredunord-VirtualBox:~/q07$ python3.8 merge_sort.py
> /home/lumieredunord/q07/merge_sort.py(3)merge()
-> result = []
(Pdb) p left
[3]
(Pdb) p right
[1]
(Pdb) n
> /home/lumieredunord/q07/merge_sort.py(4)merge()
-> i = j = 0
(Pdb) n
> /home/lumieredunord/q07/merge_sort.py(5)merge()
-> while i < len(left) and j < len(right):
(Pdb) n
> /home/lumieredunord/q07/merge_sort.py(6)merge()
-> if left[i] <= right[j]:
(Pdb) n
> /home/lumieredunord/q07/merge_sort.py(9)merge()
-> result.append(right[j]); j += 1
(Pdb) p i
0
(Pdb) p j
0
(Pdb) n
> /home/lumieredunord/q07/merge_sort.py(5)merge()
-> while i < len(left) and j < len(right):
(Pdb) n
> /home/lumieredunord/q07/merge_sort.py(10)merge()
-> return result+ left[i:] + right[j:]
(Pdb) p i
0
(Pdb) p j
1
(Pdb) p result
[1]
(Pdb) n
--Return--
> /home/lumieredunord/q07/merge_sort.py(10)merge()->[1, 3]
-> return result+ left[i:] + right[j:]
(Pdb) p i
0
(Pdb) pj
*** NameError: name 'pj' is not defined
(Pdb) p j
1
(Pdb) p result
[1]
(Pdb) n
--Return--
> /home/lumieredunord/q07/merge_sort.py(15)merge_sort()->[1, 3]
-> return merge(merge_sort(a[:m]), merge_sort(a[m:]))
(Pdb) n
--Call--
> /home/lumieredunord/q07/merge_sort.py(12)merge_sort()
-> def merge_sort(a):
(Pdb) q

```

```
def merge(left, right):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i]); i += 1
        else:
            result.append(right[j]); j += 1
    return result + left[i:] + right[j:]

def merge_sort(a):
    if len(a) <= 1: return a
    m = len(a) // 2
    return merge(merge_sort(a[:m]), merge_sort(a[m:]))

print(merge_sort([3, 1, 4, 1, 5, 9, 2, 6]))
```

```
lumieredunord@lumieredunord-VirtualBox:~/q07$ nano test_merge.py
lumieredunord@lumieredunord-VirtualBox:~/q07$ python3.8 -m pytest
=====
platform linux -- Python 3.8.0, pytest-3.3.2, py-1.5.2, pluggy-0.6.0
rootdir: /home/lumieredunord/q07, inifile:
collected 2 items

test_merge.py ..

=====
lumieredunord@lumieredunord-VirtualBox:~/q07$ python3.8 -m ruff check . --fix
Found 1 error (1 fixed, 0 remaining).
```

1.4 第 8 题

1.4.1 (1) 实验要求:

- 生成 words.txt, 使用 time 运行原程序 2 次并记录耗时。
- 使用 cProfile -s cumulative 运行一次, 找出主要耗时位置。
- 在不改变最终 count 的前提下优化程序; 不得写死 1000。
- 使用相同输入再运行 2 次, 计算优化前后中位数和加速比。

1.4.2 (2) 代码及运行结果:

```
import random

random.seed(20260831)
vocab = [f"w{i}" for i in range(1000)]
with open("words.txt", "w", encoding="utf-8") as f:
    f.write(" ".join(random.choice(vocab) for _ in range(300000)))
```

```

words = open("words.txt",encoding="utf-8").read().split()
unique=[]
for word in words:
    if word not in unique:
        unique.append(word)
print("count=",len(unique))

```

```

lumieredunord@lumieredunord-VirtualBox:~/q08$ time python3.8 wordfreq.py
count= 1000

real    0m0.968s
user    0m0.482s
sys     0m0.483s
lumieredunord@lumieredunord-VirtualBox:~/q08$ time python3.8 wordfreq.py
count= 1000

real    0m1.011s
user    0m0.975s
sys     0m0.028s
lumieredunord@lumieredunord-VirtualBox:~/q08$ python3.8 -m cProfile -s cumulative wordfreq.py
count= 1000
1012 function calls in 0.963 seconds

Ordered by: cumulative time

ncalls  tottime  percall  cumtime  percall filename:lineno(function)
1       0.000   0.000   0.963    0.963 {built-in method builtins.exec}
1       0.952   0.952   0.963    0.963 wordfreq.py:1(<module>)
1       0.009   0.009   0.009    0.009 {method 'split' of 'str' objects}
1       0.001   0.001   0.001    0.001 {method 'read' of 'io.TextIOWrapper' objects}
1       0.000   0.000   0.000    0.000 codecs.py:319(decode)
1       0.000   0.000   0.000    0.000 {built-in method _codecs.utf_8_decode}
1       0.000   0.000   0.000    0.000 {built-in method builtins.print}
1000    0.000   0.000   0.000    0.000 {method 'append' of 'list' objects}
1       0.000   0.000   0.000    0.000 {built-in method io.open}
1       0.000   0.000   0.000    0.000 {built-in method builtins.len}
1       0.000   0.000   0.000    0.000 codecs.py:309(__init__)
1       0.000   0.000   0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
1       0.000   0.000   0.000    0.000 codecs.py:260(__init__)

```

```

words = open("words.txt",encoding="utf-8").read().split()
unique_set=set()
for word in words:
    unique_set.add(word)
unique=list(unique_set)
print("count=",len(unique))

```

```

lumieredunord@lumieredunord-VirtualBox:~/q08$ time python3.8 wordfreq.py
count= 1000

real    0m0.054s
user    0m0.027s
sys     0m0.027s
lumieredunord@lumieredunord-VirtualBox:~/q08$ time python3.8 wordfreq.py
count= 1000

real    0m0.052s
user    0m0.051s
sys     0m0.000s

```

对于 real:

优化前中位数: $(0.968 + 1.011) \div 2 = 0.9895$ s

优化后中位数: $(0.054 + 0.052) \div 2 = 0.053$ s

加速比: $0.9895 \div 0.053 \approx 18.67$ 倍

2 课后练习

1. 你可能见过像 `cmd --flag -- --notafalg` 这样的命令。这里的 `--` 是一个特殊参数，它告诉程序后面不要再继续解析选项 (flag) 了。也就是说，`--` 后面的所有内容都会被当作位置参数 (positional argument)。这有什么用？试着运行 `touch -- -myfile`，然后在不使用 `--` 的情况下把它删掉。

答：`--` 告诉程序，后面不再解析命令选项，全部当做文件名。文件名以 `-` 开头必须用它，否则会把文件名当成命令参数。

```
lumieredunord@lumieredunord-VirtualBox:~$ touch -- -myfile
lumieredunord@lumieredunord-VirtualBox:~$ rm -myfile
rm: 不适用的选项 -- m
Try 'rm ./-myfile' to remove the file '-myfile'.
Try 'rm --help' for more information.
lumieredunord@lumieredunord-VirtualBox:~$ rm -- -myfile
```

2. 阅读 `man ls`，写出一个 `ls` 命令，让它按下面的方式列出文件：

- 包含所有文件，包括隐藏文件
- 文件大小以易读格式显示（例如 454M，而不是 454279954）
- 按最近修改时间排序
- 输出带颜色

答：

```
lumieredunord@lumieredunord-VirtualBox:~$ ls -alht --color=auto
总用量 196K
drwxr-xr-x 24 lumieredunord lumieredunord 4.0K 9月 6 22:01 .
-rw-r----- 1 lumieredunord lumieredunord 2.0K 9月 6 21:59 .ICEauthority
-rw-r----- 1 lumieredunord lumieredunord 7.3K 8月 31 17:14 .bash_history
drwxr-xr-x 3 lumieredunord lumieredunord 4.0K 8月 31 17:13 q08
drwxr-xr-x 4 lumieredunord lumieredunord 4.0K 8月 31 16:41 q07
drwx----- 13 lumieredunord lumieredunord 4.0K 8月 31 16:11 .config
drwx----- 15 lumieredunord lumieredunord 4.0K 8月 31 16:03 .cache
drwxr-xr-x 5 lumieredunord lumieredunord 4.0K 8月 31 15:43 q06
-rw-r--r-- 1 lumieredunord lumieredunord 176 8月 31 15:25 .wget-hsts
drwx----- 5 lumieredunord lumieredunord 4.0K 8月 31 15:05 .local
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 31 14:03 q05
-rw-r----- 1 lumieredunord lumieredunord 5 8月 31 13:43 .vboxclient-vmvga-session-tty2-control.pid
-rw-r----- 1 lumieredunord lumieredunord 5 8月 31 13:43 .vboxclient-hostversion-tty2-control.pid
-rw-r----- 1 lumieredunord lumieredunord 5 8月 31 13:43 .vboxclient-draganddrop-tty2-control.pid
-rw-r----- 1 lumieredunord lumieredunord 5 8月 31 13:43 .vboxclient-seamless-tty2-control.pid
-rw-r----- 1 lumieredunord lumieredunord 5 8月 31 13:43 .vboxclient-clipboard-tty2-control.pid
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 31 00:44 latex_practice
-rw-r--r-- 1 lumieredunord lumieredunord 19 8月 31 00:04 .gitignore_global
-rw-r--r-- 1 lumieredunord lumieredunord 173 8月 31 00:02 .gitconfig
drwxr-xr-x 35 lumieredunord lumieredunord 20K 8月 30 23:43 git
-rw-r--r-- 1 lumieredunord lumieredunord 546 8月 30 23:25 all.txt
-rw-r--r-- 1 lumieredunord lumieredunord 60 8月 30 23:23 err.txt
-rw-r--r-- 1 lumieredunord lumieredunord 486 8月 30 23:23 out.txt
-rwxr-xr-x 1 lumieredunord lumieredunord 125 8月 30 23:23 file_check.sh
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 17:12 q04
drwxr-xr-x 3 lumieredunord lumieredunord 4.0K 8月 24 16:34 q03
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 16:20 q02
drwxr-xr-x 4 lumieredunord lumieredunord 4.0K 8月 24 15:55 q01
-rw-r--r-- 1 lumieredunord lumieredunord 0 8月 24 15:39 .sudo_as_admin_successful
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 公共的
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 模板
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 视频
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 图片
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 文档
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 下载
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 音乐
drwxr-xr-x 2 lumieredunord lumieredunord 4.0K 8月 24 15:30 桌面
drwx----- 3 lumieredunord lumieredunord 4.0K 8月 24 15:30 .gnupg
-rw-r--r-- 1 lumieredunord lumieredunord 220 8月 24 15:19 .bash_logout
-rw-r--r-- 1 lumieredunord lumieredunord 3.7K 8月 24 15:19 .bashrc
-rw-r--r-- 1 lumieredunord lumieredunord 8.8K 8月 24 15:19 examples.desktop
-rw-r--r-- 1 lumieredunord lumieredunord 807 8月 24 15:19 .profile
drwxr-xr-x 3 root root 4.0K 8月 24 15:19 ..
```

3. 进程替换 `<(command)` 可以让你把一个命令的输出当成文件来用。试着配合 `diff` 和进程替换，比较 `printenv` 与 `export` 的输出。它们为什么不一样？（提示：可以试试 `diff <(printenv | sort) <(export | sort)`）

答：`printenv` 只输出环境变量；`export` 输出所有被 `export` 标记的变量，还会输出 `shell` 函数定义。

```
lumi@lumi-VirtualBox:~$ diff <(printenv | sort) <(export | sort)
1,52c1,52
< CLUTTER_IM_MODULE=xim
< COLORTERM=truecolor
< DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
< DESKTOP_AUTOSTART_ID=105a5ea5b45fdd47e6178870314049032300000013380007
< DESKTOP_SESSION=ubuntu
< DISPLAY=:0
< GDMSESSION=ubuntu
< GNOME_DESKTOP_SESSION_ID=this-is-deprecated
< GNOME_SHELL_SESSION_MODE=ubuntu
< GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/9088a9fa_2396_4119_b5cf_f100a645dc96
< GNOME_TERMINAL_SERVICE=:1.65
< GPG_AGENT_INFO=/run/user/1000/gnupg/S.gpg-agent:0:1
< GTK_IM_MODULE=ibus
< GTK_MODULES=gail:atk-bridge
< HOME=/home/lumi
< IM_CONFIG_PHASE=2
< LANGUAGE=zh_CN:zh
< LANG=zh_CN.UTF-8
< LESSCLOSE=/usr/bin/lesspipe %s %s
< LESSOPEN=| /usr/bin/lesspipe %s
< LOGNAME=lumi
< LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.Z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcqt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.d=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.
< PATH=/home/lumi/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
< PWD=/home/lumi
< QT4_IM_MODULE=xim
< QT_ACCESSIBILITY=1
< QT_IM_MODULE=xim
< SESSION_MANAGER=local/lumi-VirtualBox:@/tmp/.ICE-unix/1338,unix/lumi-VirtualBox:/tmp/.ICE-unix/1338
< SHELL=/bin/bash
< SHLVL=1
< SSH_AGENT_PID=1434
< SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
< TERM=xterm-256color
< TEXTDOMAINDIR=/usr/share/locale/
< TEXTDOMAIN=im-config
< USER=lumi
< USERNAME=lumi
< _=/usr/bin/printenv
< VTE_VERSION=5202
< WINDOWPATH=2
< XAUTHORITY=/run/user/1000/gdm/Xauthority
< XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
< XDG_CURRENT_DESKTOP=ubuntu:GNOME
< XDG_DATA_DIRS=/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop
< XDG_MENU_PREFIX=gnome-
< XDG_RUNTIME_DIR=/run/user/1000
< XDG_SEAT=seat0
< XDG_SESSION_DESKTOP=ubuntu
< XDG_SESSION_ID=2
< XDG_SESSION_TYPE=x11
< XDG_VTNR=2
< XMODIFIERS=@im=ibus
---
> declare -x CLUTTER_IM_MODULE="xim"
> declare -x COLORTERM="truecolor"
> declare -x DBUS_SESSION_BUS_ADDRESS="unix:path=/run/user/1000/bus"
> declare -x DESKTOP_AUTOSTART_ID="105a5ea5b45fdd47e6178870314049032300000013380007"
> declare -x DESKTOP_SESSION="ubuntu"
> declare -x DISPLAY=":0"
> declare -x GDMSESSION="ubuntu"
> declare -x GNOME_DESKTOP_SESSION_ID="this-is-deprecated"
> declare -x GNOME_SHELL_SESSION_MODE="ubuntu"
> declare -x GNOME_TERMINAL_SCREEN="/org/gnome/Terminal/screen/9088a9fa_2396_4119_b5cf_f100a645dc96"
> declare -x GNOME_TERMINAL_SERVICE=":1.65"
> declare -x GPG_AGENT_INFO="/run/user/1000/gnupg/S.gpg-agent:0:1"
> declare -x GTK_IM_MODULE="ibus"
```

4. 写两个 `bash` 函数 `marco` 和 `polo`，行为如下：每次执行 `marco` 时，都要以某种方式保存当前工作目录；之后无论你切到哪个目录，只要执行 `polo`，它都应该把你 `cd` 回执行 `marco` 时所在的目录。为了方便调试，你可以把代码写进 `marco.sh`，然后通过执行 `source marco.sh` 把这些定义重新加载到当前 `shell`。

答：

```
marco(){
    export MARCO_SAVE_DIR="$PWD"
    echo "$MARCO_SAVE_DIR has been saved."
}

polo(){
    if [ -n "$MARCO_SAVE_DIR" ]; then
        cd "$MARCO_SAVE_DIR"
        echo "Return to: $PWD"
    else
        echo "Error!"
    fi
}
```

```
lumieredunord@lumieredunord-VirtualBox:~$ nano marco.sh
lumieredunord@lumieredunord-VirtualBox:~$ source marco.sh
lumieredunord@lumieredunord-VirtualBox:~$ marco
/home/lumieredunord has been saved.
lumieredunord@lumieredunord-VirtualBox:~$ cd /etc
lumieredunord@lumieredunord-VirtualBox:/etc$ polo
Return to: /home/lumieredunord
```

5. 假设你有一个很少失败的命令。为了调试它，你需要把它的输出保存下来，但等到一次失败运行可能会很耗时。写一个 bash 脚本，不断运行下面这个脚本直到它失败为止，并把标准输出和标准错误分别保存到文件里，最后把结果打印出来。如果你还能顺便报告它运行了多少次才失败，就加分。

```
#!/sr/bin/env bash
n=$(( RANDOM % 100 ))
if [[ n -eq 42 ]]; then
    echo "Something went wrong"
    >&2 echo "The error was using magic numbers"
    exit 1
fi
echo "Everything went according to plan"
```

答：

```
#!/usr/bin/env bash

n=$(( RANDOM % 100 ))

if [[ $n -eq 42 ]]; then
    echo "Something went wrong" >&2
    echo "The error was using magic numbers" >&2
    exit 1
fi

echo "Everything went according to plan"
```

```
#!/usr/bin/env bash

count=0

>stdout.txt
>stderr.txt

while true
do
    count=$((count + 1))
    ./rand_fail.sh >> stdout.txt 2>> stderr.txt
    ret=$?
    if [ $ret -ne 0 ]; then
        echo "Error!Running times:$count"
        echo "===stdout.txt==="
        cat stdout.txt
        echo "===stderr.txt==="
        cat stderr.txt
        break
    fi
done
```

```
lumieredunord@lumieredunord-VirtualBox:~$ nano rand_fail.sh
lumieredunord@lumieredunord-VirtualBox:~$ chmod +x rand_fail.sh
lumieredunord@lumieredunord-VirtualBox:~$ nano loop_until_fail.sh
lumieredunord@lumieredunord-VirtualBox:~$ chmod +x loop_until_fail.sh
lumieredunord@lumieredunord-VirtualBox:~$ ./loop_until_fail.sh
Error!Running times:256
===stdout.txt===
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
```

```
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
Everything went according to plan
===stderr.txt===
Something went wrong
The error was using magic numbers
```

6. 调试排序算法：下面的伪代码实现了归并排序，但存在一个错误。请用你选择的编程语言来实现该算法，然后使用调试器（如 gdb、lldb、pdb，或是你所用集成开发环境中的调试工具）来找出并修复这个错误。

```
function merge_sort(arr):
    if length(arr) <= 1:
        return arr
    mid = length(arr) / 2
    left = merge_sort(arr[0..mid])
    right = merge_sort(arr[mid..end])
    return merge(left, right)

function merge(left, right):
    result = []
    i = 0, j = 0
    while i < length(left) AND j < length(right):
        if left[i] <= right[j]:
            append result, left[i]
            i = i + 1
        else:
            append result, right[i]
            j = j + 1
    append remaining elements from left and right
    return result
```

答：

```

def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

def merge(left, right):
    import pdb
    pdb.set_trace()
    result = []
    i = 0
    j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    result.extend(left[i:])
    result.extend(right[j:])
    return result

test_arr = [3, 1, 4, 1, 5, 9, 2, 6]
res = merge_sort(test_arr)
print(res)

```

```

lumieredunord@lumieredunord-VirtualBox:~$ nano merge_sort.py
lumieredunord@lumieredunord-VirtualBox:~$ python3 merge_sort.py
> /home/lumieredunord/merge_sort.py(12)merge()
-> result = []
(Pdb) p left
[3]
(Pdb) p right
[1]
(Pdb) p i
*** NameError: name 'i' is not defined
(Pdb) p j
*** NameError: name 'j' is not defined
(Pdb) n
> /home/lumieredunord/merge_sort.py(13)merge()
-> i = 0
(Pdb) n
> /home/lumieredunord/merge_sort.py(14)merge()
-> j = 0
(Pdb) n
> /home/lumieredunord/merge_sort.py(15)merge()
-> while i < len(left) and j < len(right):
(Pdb) n
> /home/lumieredunord/merge_sort.py(16)merge()
-> if left[i] <= right[j]:
(Pdb) n
> /home/lumieredunord/merge_sort.py(20)merge()
-> result.append(right[i])
(Pdb) p i
0
(Pdb) p j
0
(Pdb) p right[i]
1
(Pdb) p right[j]
1
1

```

```
def merge_sort(arr):
    if len(arr) <= 1:
        return arr
    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

def merge(left, right):
    result = []
    i = 0
    j = 0
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1
    result.extend(left[i:])
    result.extend(right[j:])
    return result

test_arr = [3, 1, 4, 1, 5, 9, 2, 6]
res = merge_sort(test_arr)
print(res)
```

```
lumieredunord@lumieredunord-VirtualBox:~$ nano merge_sort.py
lumieredunord@lumieredunord-VirtualBox:~$ python3 merge_sort.py
[1, 1, 2, 3, 4, 5, 6, 9]
```

7. 使用 strace (Linux 系统) 或 dtruss (macOS 系统) 来追踪像 `ls -l` 这样的命令所发起的系统调用。它到底发出了哪些系统调用呢? 可以试着追踪一个更复杂的程序, 看看它会打开哪些文件。

答:

- `execve`: 加载运行 `ls` 程序。
- `mmap`: 加载系统 C 标准库等动态链接库。
- `openat`: 打开目标目录。
- `getdents64`: 读取目录下所有文件名。
- `fstat`: 对每一个文件获取权限、大小、修改时间 (`ls -l` 长列表的数据来源)。
- `write`: 将格式化后的列表字符串输出到终端。
- `close`: 关闭文件描述符, `exit_group` 进程退出。

```

lumieredunord@lumieredunord-VirtualBox:~$ strace ls -l
execve("/bin/ls", ["ls", "-l"], 0x7ffedf4b24e8 /* 54 vars */) = 0
brk(NULL)                                = 0x55ffb8f26000
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or d
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or d
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=78501, ...}) = 0
mmap(NULL, 78501, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7efdad708000
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or d
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_C
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20b\0\0\0\0\0
fstat(3, {st_mode=S_IFREG|0644, st_size=154832, ...}) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1,
mmap(NULL, 2259152, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3,
mprotect(0x7efdad2f0000, 2093056, PROT_NONE) = 0
mmap(0x7efdad4ef000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
mmap(0x7efdad4f1000, 6352, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or d
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\240\35\2\0\0\0
fstat(3, {st_mode=S_IFREG|0755, st_size=2030928, ...}) = 0
mmap(NULL, 4131552, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3,
mprotect(0x7efdad0c1000, 2097152, PROT_NONE) = 0
mmap(0x7efdad2c1000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
mmap(0x7efdad2c7000, 15072, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or d
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcr.so.3", O_RDONLY|O_CLOE
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \25\0\0\0\0\0
fstat(3, {st_mode=S_IFREG|0644, st_size=464824, ...}) = 0
mmap(NULL, 2560264, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3,
mprotect(0x7efdacc8000, 2097152, PROT_NONE) = 0
mmap(0x7efdaced8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or d
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libdl.so.2", O_RDONLY|O_CLOEXE
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\16\0\0\0\0\0
fstat(3, {st_mode=S_IFREG|0644, st_size=14560, ...}) = 0
mmap(NULL, 2109712, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3,
mprotect(0x7efdaca67000, 2093056, PROT_NONE) = 0
mmap(0x7efdacc66000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or d

```

```

lumieredunord@lumieredunord-VirtualBox:~$ strace ls -l > ls_out.txt 2> strace_log.txt
lumieredunord@lumieredunord-VirtualBox:~$ cat strace_log.txt
execve("/bin/ls", ["ls", "-l"], 0x7fffd08ec868 /* 54 vars */) = 0
brk(NULL)                               = 0x559fabcad000
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or directory)
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=78501, ...}) = 0
mmap(NULL, 78501, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f61aeb6a000
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20b\0\0\0\0\0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=154832, ...}) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f61aeb6800
mmap(NULL, 2259152, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f61ae72d0
mprotect(0x7f61ae752000, 2093056, PROT_NONE) = 0
mmap(0x7f61ae951000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0x7f61ae951000
mmap(0x7f61ae953000, 6352, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f61ae953000
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\240\35\2\0\0\0\0"... , 832) = 8
fstat(3, {st_mode=S_IFREG|0755, st_size=2030928, ...}) = 0
mmap(NULL, 4131552, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f61ae33c0
mprotect(0x7f61ae523000, 2097152, PROT_NONE) = 0
mmap(0x7f61ae723000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0x7f61ae723000
mmap(0x7f61ae729000, 15072, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f61ae729000
close(3)                                 = 0
access("/etc/ld.so.nohwcap", F_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre.so.3", O_RDONLY|O_CLOEXEC) = 3

```

8. 使用 `perf stat` 可以获取所选程序的基本性能统计数据。那些不同的计数器代表什么含义呢？

答：

```

#include<math.h>
#include<stdio.h>

double slow_computation(int n){
    double result = 0;
    for (int i=0;i<n;i++){
        for (int j=0;j<1000;j++){
            result += sin(i*j)*cos(i+j);
        }
    }
    return result;
}

int main(){
    double r=0;
    for (int i=0;i<100;i++){
        r += slow_computation(1000);
    }
    printf("Result:%f\n",r);
    return 0;
}

```

```

lumieredunord@lumieredunord-VirtualBox:~$ perf stat ./slow
Result:-122140.186478

Performance counter stats for './slow':

      33.39 msec task-clock           #    0.871 CPUs utilized
         4      context-switches      #    0.120 K/sec
         0      cpu-migrations        #    0.000 K/sec
        70      page-faults          #    0.002 M/sec
<not supported>      cycles
<not supported>      instructions
<not supported>      branches
<not supported>      branch-misses

    0.038324078 seconds time elapsed

    0.000000000 seconds user
    0.034590000 seconds sys

lumieredunord@lumieredunord-VirtualBox:~$ perf record -g ./slow
Couldn't record kernel reference relocation symbol
Symbol resolution may be skewed if relocation was used (e.g. kexec).
Check /proc/kallsyms permission or run as root.
Result:-122140.186478
[ perf record: Woken up 1 times to write data ]
[ perf record: Captured and wrote 0.002 MB perf.data (13 samples) ]
lumieredunord@lumieredunord-VirtualBox:~$ perf report

```

9. 带有 perf record 的配置文件。将其保存为 slow.c:

```

#include <math.h>
#include <stdio.h>

double slow_computation(int n) {
    double result = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < 1000; j++) {
            result += sin(i * j) * cos(i + j);
        }
    }
    return result;
}

int main() {
    double r = 0;
    for (int i = 0; i < 100; i++) {
        r += slow_computation(1000);
    }
    printf("Result: %f\n", r);
    return 0;
}

```

使用调试符号进行编译: `gcc -g -O2 slow.c -o slow -lm`。运行 `perf record -g ./slow`, 然后再运行 `perf report`, 这样就能知道时间被消耗在了哪些地方。

答:

Samples: 11 of event 'cpu-clock:pppH', Event count (approx.): 2750000					
Children	Self	Command	Shared	Object	Symbol
+ 27.27%	18.18%	slow	libm-2.27.so	[.]	__sin_fma
+ 27.27%	27.27%	slow	libc-2.27.so	[.]	intel_check_word.isra.0
+ 18.18%	18.18%	slow	ld-2.27.so	[.]	get_common_indeces.constprop.1
+ 18.18%	18.18%	slow	libm-2.27.so	[.]	__cos_fma
+ 9.09%	9.09%	slow	[unknown]	[k]	0xfffffffff9be989c5
+ 9.09%	0.00%	slow	[unknown]	[k]	0x312e303431323231
+ 9.09%	0.00%	slow	libc-2.27.so	[.]	__GI___libc_write
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9c00008c
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b40441a
+ 9.09%	9.09%	slow	[unknown]	[k]	0xfffffffff9c200071
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b6cc52a
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b6cc4d7
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9c000a34
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b4042e1
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b6c9f11
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b404029
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b6c923b
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9ba61994
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9c000a0f
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9ba636d3
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9c001e86
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9b4a0cfe
+ 9.09%	0.00%	slow	[unknown]	[.]	0xfffffffff9ba6b5f1

10. 一个常见的问题是：你想用来监听数据的端口已经被其他进程占用了。下面介绍如何找出那个进程：首先执行 `python -m http.server 4444`，在 4444 端口上启动一个简单的 Web 服务器。然后在另一个终端中执行 `ss -tlnp | grep 4444` 来查找那个进程，最后用 `kill <PID>` 终止它。

答：

```

lumieredunord@lumieredunord-VirtualBox: ~
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
lumieredunord@lumieredunord-VirtualBox:~$ ss -tlnp | grep 444
LISTEN 0      5          0.0.0.0:*      0.0.0.0:4444      0.0.0.0:*
users:((("python3",pid=6524,fd=3))
lumieredunord@lumieredunord-VirtualBox:~$ ss -tlnp | grep 4444 | awk '{print $6}'
' | grep -oE 'pid=[0-9]+' | grep -oE '[0-9]+'
lumieredunord@lumieredunord-VirtualBox:~$ kill $(ss -tlnp | grep 4444 | awk '{print $6}' | grep -oE 'pid=[0-9]+' | grep -oE '[0-9]+')
lumieredunord@lumieredunord-VirtualBox:~$ ss -tlnp | grep 4444
lumieredunord@lumieredunord-VirtualBox:~$

lumieredunord@lumieredunord-VirtualBox:~/FlameGraph$ python3 -m http.server 4444
Serving HTTP on 0.0.0.0 port 4444 (http://0.0.0.0:4444/) ...
已终止

```

3 解题感悟

通过学习命令行、开发工具、调试与性能分析，我体会到命令行是底层操作的基础。调试工具帮助定位代码逻辑错误，性能分析工具可以发现程序运行瓶颈。很多问题不能只靠肉眼读代码，要借助工具采集真实运行数据。同时虚拟机环境会带来采样、软件包兼容等限制，需要灵活调整方案解决实操中的各类问题。

github 链接：<https://github.com/Lumieredunord/linux-lab-report>

linux-lab-report / 课堂练习 /

Lumieredunord 系统开发工具基础作业

Name	Last commit message
..	
q01	系统开发工具基础作业
q02	系统开发工具基础作业
q03	系统开发工具基础作业
q04	系统开发工具基础作业
q05	系统开发工具基础作业
q06	系统开发工具基础作业
q07	系统开发工具基础作业
q08	系统开发工具基础作业